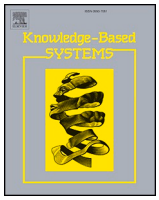
Knowledge-Based Systems 351 (2026) 116603



Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys



Improved grey wolf optimization-based artificial neural network for irrigation need prediction

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ARTICLE INFO

Keywords:
Artificial neural network
Gray wolf optimization
Irrigation
Internet of Things
Water management

ABSTRACT

Efficient irrigation management in agriculture is key to optimizing water use, ensuring optimal crop health and productivity. Existing machine learning (ML) models demonstrate insufficient classification accuracy in classifying irrigation needs for various crops, motivating new methods. This study identifies a gap in using bio-inspired optimization algorithms and ML to enhance irrigation classification. This research develops a hybrid intelligent model combining the improved grey wolf optimization (IGWO) and artificial neural networks (ANN) to classify irrigation needs for different crops. IGWO employs an adaptive weighted position update strategy that assigns greater influence to the best solution during ANN weight initialization, effectively avoiding local minima traps and providing more consistent near-optimal parameter configurations compared to standard optimization approaches. The IGWO-ANN model is trained using crucial soil and environmental variables including soil moisture, air temperature, humidity, and soil temperature. The proposed model achieves a mean accuracy of 98.72% $\pm$ 0.44%, with precision, recall, F1-score, and AUC of 93.46% $\pm$ 1.32%, 99.43% $\pm$ 1.47%, 96.35% $\pm$ 1.25%, and 0.9994 $\pm$ 0.0009 respectively, evaluated across 10 independent runs to ensure statistical reliability. Additionally, the gradient-based fine-tuning phase requires only 0.44 s, indicating computational efficiency. The proposed model is benchmarked against 10 competing models including metaheuristic variants, deep learning architectures, and ensemble methods, demonstrating superior and statistically stable classification performance. The results reveal that the proposed IGWO-ANN approach enhances prediction reliability and reduces the limitations of standalone ANN models. Thus, the proposed model can significantly reduce water loss during the irrigation process and contribute toward sustainable agriculture and efficient water resource management.

1. Introduction

Due to the increasing impacts of climate change and global population growth, the future of agriculture is at a critical stage. Water is essential for agricultural activities and for sustaining life. However, managing water for irrigation in both agriculture and domestic sectors remains a significant challenge [1]. As food demand is projected to increase substantially in the coming years, sustainable approaches are important for improving agricultural productivity and crop yields. Water scarcity is a serious problem in many regions, and since irrigation uses most of this vital resource, the need for action is urgent. Agriculture uses approximately 70% of the world's groundwater, and about 38% of irrigated land depends on it. However, heavy reliance on groundwater for intensive agriculture raises serious concerns about climate resilience, clean water availability, food security, and the ecological balance of groundwater-dependent water bodies and wetlands [2].

For these reasons, many technological research areas have started exploring ways to use water in an efficient and sustainable manner. The two fields making a big difference in this area are Artificial Intelligence (AI) and the Internet of Things (IoT) [2]. IoT is rapidly expanding due to the increasing variety of available sensor types and their decreasing costs. Nowadays, installing sensors for large-scale farms has become

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https://doi.org/10.1016/j.knosys.2026.116603
Received 12 April 2026; Received in revised form 21 June 2026; Accepted 2 July 2026
Available online 3 July 2026
0950-7051/© 2026 Published by Elsevier B.V.

easier due to advances in communication technologies, low power consumption, and simple installation methods. Sensors have become common and affordable in agriculture, and large farms are starting to recognize their many benefits, such as cost savings and remote control. Examples of sensors used in agriculture include temperature, stem width, soil moisture, and soil salinity [2]. However, practical IoT deployments in agricultural environments introduce several challenges, including sensor noise, missing or incomplete readings due to connectivity failures, and constraints related to edge-device computational capacity, power consumption, and real-time inference requirements. Addressing these challenges is essential for ensuring the robustness and scalability of AI-based irrigation classification systems in real-world field deployments.

On the other hand, AI has become a widely adopted research field and researchers are continuously developing new models and algorithms. With its advancement, it is now possible in agriculture to create data-based programs that make intelligent decisions about irrigation, thus reducing water waste. Furthermore, many AI algorithms can make decisions based on data collected from sensors. For this purpose, both machine learning (ML) methods such as support vector machines and naïve bayes (NB and more complex deep learning methods such as artificial neural networks (ANN) can be used to classify irrigation requirements for different crops. However, these methods face several challenges in predicting irrigation. For instance, traditional ML algorithms have limited classification accuracy [3], scalability issues when crop types, soil conditions, environment factors, or irrigation conditions differ [4], massive data requirements [5], black box behavior, uncertainty in simulation results [6], and challenges in automatically identifying and fine-tuning features for desired outcomes [7]. Moreover, bias and overfitting remain common issues in ML methods. The main causes of overfitting are noisy data, insufficient training data, and model complexity [8].

Conversely, ANN has emerged as one of the most widely used DL models for irrigation need prediction. ANN has gained significant attention due to its reliability and ability to maximize predictive accuracy through the backpropagation algorithm during the training phase [5,9]. ANN can learn complex patterns and relationships within data by iteratively adjusting weights using backpropagation, enabling effective modelling of both linear and nonlinear interactions between input and output variables [10]. Furthermore, their capacity to handle high-dimensional data makes them particularly suitable for complex prediction tasks [11]. Consequently, the validity of using ANN models for classifying irrigation needs has been widely investigated in the literature [3,5,12,13]. However, most reported studies on irrigation need prediction rely on standalone data intelligent algorithms without incorporating parameter tuning or optimization mechanisms. These models typically explore a large search space during dataset training, which may limit their ability to achieve globally optimal solutions. To address these limitations, recent studies have demonstrated that integrating AI models with biologically inspired optimization techniques leads to improved parameter tuning and enhanced model performance [14–17]. In particular, the hybridization of ANN with metaheuristic optimization algorithms has been shown to significantly improve prediction accuracy in various agricultural water management applications [15,16,18–20].

Among various bio-inspired optimization algorithms, the Grey Wolf Optimizer (GWO) has recently emerged as an effective global optimization technique for solving complex optimization problems [21]. More recently, GWO has been combined with ANN models to enhance their prediction performance and robustness [14,22,23]. GWO imitates the natural hunting behavior and social hierarchy of grey wolves. Similar to other meta-heuristic algorithms, GWO starts the optimization process by randomly generating a group of candidate solutions. During each iteration, the three best solutions are designated as wolves (alpha, beta, and delta), which guide the other wolves toward promising regions in the solution space. The remaining omega wolves follow the three best wolves to explore and identify better solutions. As a meta-heuristic tool, GWO aims to search for globally optimal parameters that enhance the predictive capability of AI models efficiently. Studies around the integration of the ANN model with GWO are yet to be reported for the prediction of soil moisture estimation [14,24,25], reference evapotranspiration (ETo) prediction [25–27], soil salinity prediction [10], crop yield prediction [28], and irrigation canal system simulation [29]. While hybrid metaheuristic-ANN approaches have been investigated in these agricultural prediction tasks, their application to irrigation needs classification remains underexplored in the existing literature.

The objective of this research is outlined as follows. (1) To implement and evaluate an Improved Grey Wolf Optimization combined with the Artificial Neural Network model (i.e., IGWO-ANN) for classifying the irrigation requirements of different crops. (2) The performance of the proposed model is validated against existing state-of-the-art methods including metaheuristic variants, deep learning architectures, and ensemble methods to demonstrate its predictive accuracy, statistical stability, and computational efficiency in irrigation classification. The remainder of the article is split into the following sections. Section 2 describes the materials and methods. Section 3 outlines the prerequisites of the proposed approach. Section 4 details the experimental setup and the ensuing results. The results and discussion are covered in Section 5. Finally, Section 5 concludes the article with a summary of findings and future work directions.

2. Materials and methods

2.1. Data collections

The data used in this study was collected by IoT sensors and obtained from the Mendeley Data repository (https://data.mendeley.com/datasets/67gkrzbwrr/1). It was collected through IoT devices and sensors deployed across agricultural fields in Iraq over one week. This dataset encompasses several variables including time, crop type, air temperature, soil moisture, air humidity, soil temperature, and irrigation, as shown in Table 1. The irrigation and crop type columns are binary, the time column represents the date and time of recording, and all other columns are continuous. The dataset contains only two crop values: 1 (paddy) and 2 (barley). Based on the recorded values, the irrigation column indicates whether irrigation is required or not. Therefore, this is the column that the model predicts for new readings after learning patterns from the training set.

2.2. Data cleaning and pre-processing

The process of ‘‘cleaning’’ data is required as part of the analysis process for the development of an improved model. Depending on the type of data and its primary issues, a variety of methods exist for managing this process. There are a few missing values in this dataset, and the scales of the columns vary, both of which have an impact on the performance of the model and must be resolved before proceeding.

- Handling missing values: The dataset contains both numeric and categorical data types, and each requires a distinct method for handling missing values. During data preprocessing, these were

Table 1
Exploratory of the dataset.

Column Name	Column Type	Description
time	Datetime	Time of recording data
crop type	Integer	1 (paddy) / 2 (barley)
soil temperature	Float	
soil moisture	Float	
humidity	Float	
temperature	Float	
irrigation	Boolean	Target column

systematically excluded from the dataset, as only a small number of samples were missing. This approach avoids introducing bias or distortion that may arise from value imputation and ensures that only complete and valid samples are used for subsequent training and evaluation of the proposed model. It is acknowledged that broader IoT deployment considerations, including sensor noise robustness, edge-device implementation, power consumption, and real-time inference constraints, were not empirically evaluated in this study due to the use of a secondary dataset. These aspects represent important directions for future investigation and will be addressed in subsequent work involving real-world field deployments.

- Data Scaling: The data scaling technique is employed to rescale feature values into a standardized range before being provided to the learning model, which helps improve numerical stability and training efficiency. Min–Max normalization is applied in the proposed IGWO–ANN-based irrigation classification framework, motivated by prior findings [3] that demonstrated its superior performance over other scaling methods for the same dataset. This method scales all input features into the range [0,1], ensuring that no single feature dominates the learning process due to scale differences. To ensure methodological rigor and prevent data leakage, the Min–Max scaler was fitted exclusively on the training data, and the resulting scaling parameters (minimum and maximum values) were subsequently applied to the testing data without re-fitting. This ensures that the normalization process remains entirely independent of the test set, preserving the integrity of the evaluation and ensuring that the reported performance metrics reflect a true generalization of the model. The scaling operation is defined as follows:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1a}$$

Where X represents the original feature value, and X_{min} and X_{max} denote the minimum and maximum values of each feature, respectively.

3. Prerequisite

The process of predicting irrigation needs relies on datasets encompassing distinct environmental and historical patterns, which require models that can identify these patterns and trends. Common irrigation prediction methods include adaptive techniques like ML models and autoregressive approaches. In ML, a model depends on the type of learning algorithm, the number of layers, the number of neurons, and the associated weights. The conceptual design of ANN models is centered around two primary components: weight optimization and architectural structure. The architectural structure encompasses the specific design and configuration of the neural network. Meanwhile, weight optimization is about fine-tuning the weights within the model structure [30].

3.1. Gray wolf optimization (GWO)

GWO is an evolutionary algorithm recommended by Mirjalili et al., [21]. It is inspired by the hunting behavior and social hierarchy of grey wolves, which live in small packs of 5–12 wolves. These packs are organized into four main groups namely alpha (α), beta (β), delta (δ), and omega (ω). The alpha wolves are male and female. They are the dominant members and are responsible for all decisions, including hunting and daily activities such as sleeping and waking, and all other members must follow their commands. Although they are not always the strongest, they are the most capable of managing the group. The beta wolves are subordinate to the alpha and assist them in decision-making. They perform the alpha's commands across the pack and provide feedback, making them the most suitable candidates to take over the alpha role if it is lost. The delta wolves have duties such as hunters, scouts, elders, sentinels, and caretakers. They follow the beta and alpha wolves but have authority over the omega wolves. Their main duties include guarding the territory and ensuring the safety of the group. The omega wolves are placed at the bottom of the dominance hierarchy. They serve as victims who must always surrender to all other wolves and are the last to eat. Despite this, they play an important role in maintaining group harmony and caring for younger wolves. The hunting process of grey wolves has three main stages: hunting, encircling, and attacking the prey [27,31]. During encircling, wolves update their positions around the prey using the following equations:

$$D = \left| C \cdot X_{p(t)} - X(t) \right| \tag{1}$$

$$X(t+1) = X(t) - A \cdot D \tag{2}$$

Where $X(t)$ and $X(t+1)$ represent the current and updated positions of a gray wolf, respectively, $X_{p(t)}$ represents the position of the prey, and A and C are coefficient vectors defined as:

$$A = 2a \cdot r_1 - a, \; C = 2 \cdot r_2 \tag{3}$$

Here, a is a linearly decreasing parameter controlling the balance between exploitation and exploration, and r_1 and r_2 are random vectors uniformly distributed in the range [0,1].

The hunting process is guided by the three best candidate solutions, namely the alpha, beta, and delta wolves. Instead of relying on a single prey position, the wolves update their positions based on these three leading solutions as follows:

$$\begin{aligned} X_1 &= X_{\alpha(t)} - A_1 \cdot D_\alpha \\ X_2 &= X_{\beta(t)} - A_2 \cdot D_\beta \end{aligned} \tag{4}$$

$$\begin{aligned} X_3 &= X_{\delta(t)} - A_3 \cdot D_\delta \\ X(t+1) &= \frac{(X_1 + X_2 + X_3)}{3} \end{aligned} \tag{5}$$

This mechanism facilitates the convergence of wolves toward promising regions within the search space. The parameter a is gradually reduced over iterations, which enables the algorithm to transition from global exploration to local exploitation, ultimately leading to convergence toward an optimal solution. The movement behavior of gray wolves during the optimization process is illustrated in Fig. 1 [32].

3.2. ANN

ANNs are computer models inspired by the function and structure of biological neural networks, specifically the human brain. They consist of

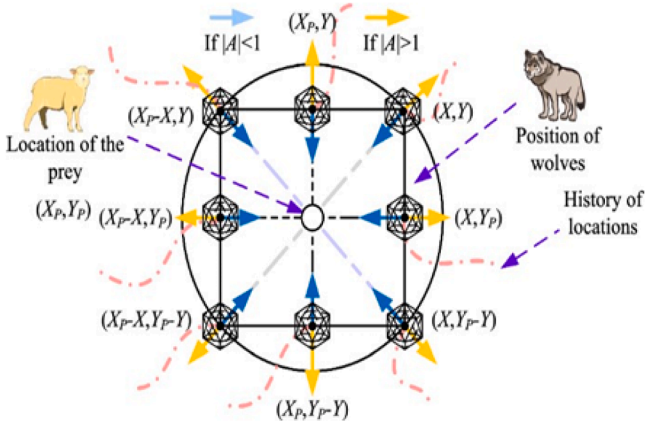


Fig. 1. Movement of gray wolves from the starting position toward the forward position during the hunting process in GWO (Reproduced with permission).

interconnected processing units, known as artificial neurons or nodes, which work together to perform specific tasks [30]. ANN processes data by passing it through a series of layers, where each connection is assigned a numerical weight that determines its importance. The process starts at the input layer, moves through hidden layers, and ends at the output layer. In a Multi-Layer Perceptron (MLP) using supervised learning, the output is checked against the target value to calculate the error signal. This signal is then used to adjust the interconnected weights between neurons, allowing them to evolve throughout the training process [33]. To optimize the network's performance, hyperparameters such as activation functions, learning rate, neurons per layer, epochs, and the number of hidden layers must be carefully adjusted. Initially, the learning rate, biases, and weights are set to values between 0 and 1. Input data is first normalized and processed, followed by the calculation of the activation functions and weighted sum within the hidden layers. The error between predictions and ground-truth labels is then determined using an appropriate loss function like Cross-Entropy for classification and Mean Squared Error for regression. During backpropagation in a network with two hidden layers, the weights are updated in reverse order, starting with the second layer and then the first. As shown in Fig. 2, this process repeats for several epochs until the model converges.

The architecture of the ANN employed in this study comprises three fully connected hidden layers with 16, 32, and 64 neurons, respectively. The Rectified Linear Unit (ReLU) activation function is applied in all three hidden layers, as it effectively addresses the vanishing gradient problem and accelerates convergence [3]. Since this study focuses on binary classification, the output layer consists of a single neuron employing the Sigmoid activation function to compute the final classification probability. A Dropout layer is incorporated before the output layer, meaning the architecture supports regularization but no dropout was actively applied following empirical tuning. All network weights and biases are initialized using PyTorch's default Kaiming Uniform initialization strategy, which is well-suited for layers employing ReLU activations. The model is trained using full-batch training, wherein the entire training dataset is utilized in each iteration. A manual weighted Binary Cross-Entropy loss function with a minority class weight of 2.2 is adopted to address the 5:1 class imbalance between the no-irrigation and irrigation classes, and the Adam optimizer with a learning rate of 0.004 is employed to update the model parameters during the gradient-based fine-tuning phase. No early stopping criterion was applied. Instead, the training was conducted for a fixed number of 100 epochs in the gradient-based fine-tuning phase, following 150 iterations of IGWO-based weight initialization in the global search phase. The key hyperparameters were determined through empirical evaluation. The learning rate of 0.004 and training duration of 100 epochs were selected based on convergence behavior observed during preliminary experiments, while the class weight value of 2.2 was determined empirically to address the class imbalance. Regarding reproducibility, the train/test split was performed using a fixed random state of 42, ensuring fully deterministic data partitioning. The stochastic components of the IGWO algorithm were intentionally left unseeded to reflect the natural stochastic behavior of metaheuristic optimization, and all reported results represent the mean ± standard deviation across 10 independent runs to provide statistically robust performance characterization. Fig. 3 illustrates the architecture of the ANN utilized in this study.

3.3. Improved Grey Wolf Neural Network Optimizer (IGWO-ANN)

The integration of the Improved Grey Wolf Optimizer (IGWO) with the ANN demonstrates a considerable enhancement in the proposed approach. The purpose of hybridizing IGWO and ANN is to identify the most suitable ANN model while minimizing the drawbacks of the standard backpropagation method. The proposed model overcomes the

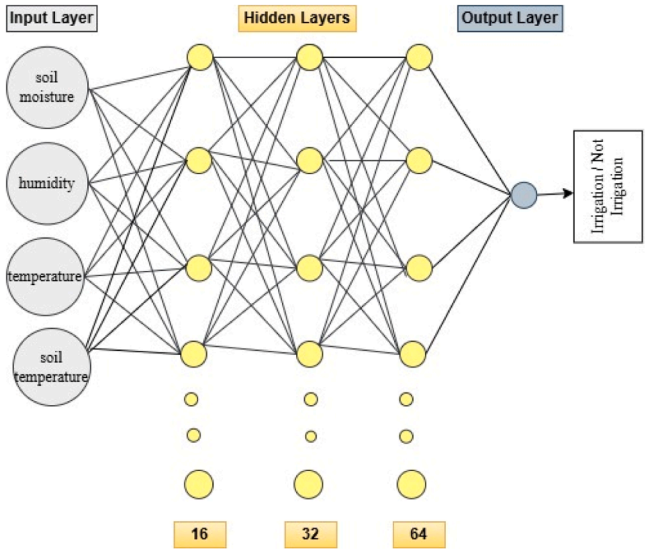


Fig. 3. ANN architecture.

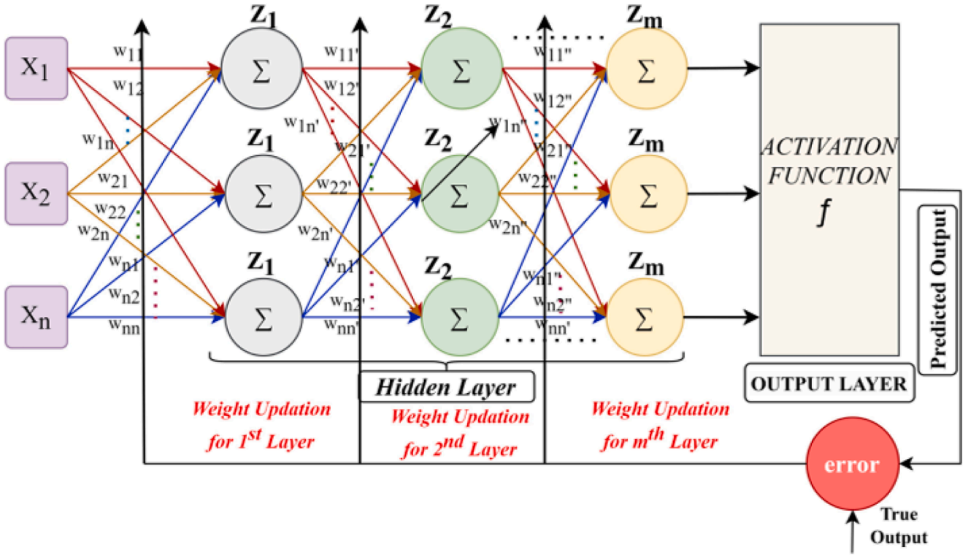


Fig. 2. Structure of ANN (Reproduced with Permission) [30].

inherent limitations of standalone techniques through a two-stage optimization process. In the first stage, IGWO performs a global search to determine the optimal initial weights and biases for the ANN. During this phase, weights and biases are evaluated as vectors, and their suitability is assessed by minimizing the weighted Binary Cross-Entropy loss between the model's predictions and ground-truth labels. In the second stage, the backpropagation algorithm utilizing the Adam optimizer fine-tunes these IGWO-derived weights through local search to reach the global optimum. The primary rationale for this hybridization is to mitigate the local minima entrapment and slow convergence often associated with the standard backpropagation algorithm when relying on random weight initialization.

The IGWO implementation employs a weighted position update strategy that distinguishes it from standard GWO. Rather than assigning equal weights of (1/3, 1/3, 1/3) to the alpha, beta, and delta leaders as in standard GWO, IGWO assigns adaptive influence weights that give progressively greater authority to the best solution. The detailed IGWO parameter configuration is summarized in Table 2.

To empirically isolate the specific contribution of each component within the proposed approach, an ablation analysis was conducted by examining the role of four key factors: IGWO-based initialization, weighted loss function, Adam optimizer, and architectural design. The analysis confirms that the weighted Binary Cross-Entropy loss function is a critical contributor to overall model performance, as without it the model would be biased toward the majority no-irrigation class. However, since this loss function is held constant across all compared variants, it does not account for IGWO's advantage over other optimization strategies. The Adam optimizer and the three-layer architecture provide moderate contributions to overall performance but are likewise identical across all variants and therefore do not explain the performance differences between optimization strategies. The primary performance advantage of IGWO is attributed to its optimized initialization mechanism. The weighted position update strategy gives greater influence to the alpha wolf, leading to more promising starting points for backpropagation. This is evidenced by IGWO achieving the lowest variance of ±0.44% across 10 independent runs compared to Standard GWO (±0.54%), Adaptive GWO (±0.60%), and Chaotic GWO (±1.55%), confirming that IGWO provides more consistent and near-optimal weight configurations that enable faster and more stable convergence during the gradient-based fine-tuning phase. The IGWO-initialized weights serve as the starting point for backpropagation, improving the overall effectiveness of the local search to find a more robust model. Once trained, the IGWO-ANN classifies irrigation needs to assist in effective water resource management, taking five input features such as crop type, soil moisture, air humidity, air temperature, and soil temperature to determine irrigation requirements. Fig. 4 illustrates the proposed IGWO-ANN optimization process.

Table 3 presents the operational procedure of the proposed IGWO-ANN model employed for irrigation classification. The process begins by loading the irrigation dataset and removing samples containing missing values, followed by feature normalization using Min-Max scaler fitted exclusively on the training data to ensure all input features lie within the range [0, 1] without introducing data leakage. The normalized data is then split into training and testing subsets using an 80:20 ratio with a fixed random state of 42 to ensure deterministic and reproducible data partitioning. The IGWO parameters are initialized as follows: the number of search agents (wolves) is set to 60, the maximum number of iterations is set to 150, the initial population positions are uniformly sampled from the range [−1, 1], and the adaptive influence weights are set as w1 = 0.6 to 0.5 for the alpha wolf, w2 = 0.3 for the beta wolf, and w3 as the residual for the delta wolf. Multiple candidate weight vectors representing the ANN parameters are randomly generated within this range. During each iteration of the IGWO process, every candidate weight vector is injected into the ANN, and a forward pass is performed using the training data to compute the fitness value based on the weighted Binary Cross-Entropy loss function with a minority class weight of 2.2 to address the 5:1 class imbalance in the dataset. The candidate solutions are ranked according to their fitness values, and the alpha, beta, and delta wolves are identified as the three best solutions. Subsequently, the positions of all candidate weight vectors are updated using the IGWO weighted position update equation $W_i = w1 \times X1 + w2 \times X2 + w3 \times X3$, which assigns greater influence to the alpha wolf compared to the equal weighting strategy of Standard GWO, thereby providing more consistent and near-optimal weight configurations. This iterative optimization process continues for 150 iterations, and the weight vector corresponding to the alpha wolf is selected as the optimal initial set of ANN weights. Following the IGWO-based weight initialization stage, the ANN is initialized with these optimized weights and further fine-tuned using Adam-based backpropagation with a learning rate of 0.004 over 100 epochs, employing the weighted Binary Cross-Entropy loss function to maintain class imbalance handling throughout the fine-tuning phase. To ensure statistical reliability, the entire process is executed across 10 independent runs and all results are reported as mean ± standard deviation. Finally, the trained IGWO-ANN model is evaluated using the testing data to generate the irrigation classification results across all reported performance metrics.

Table 2
IGWO best configuration.

Parameter	Value
Number of wolves	60
Iterations	150
Initial position range	[−1, 1]
Alpha weight (w1)	0.6 → 0.5 (decreases linearly)
Beta weight (w2)	0.3
Delta weight (w3)	0.1 → 0.2 (residual, increases)
Control parameter a	Linear decrease 2 → 0
Position update	Weighted: w1 × X1 + w2 × X2 + w3 × X3

3.4. Performance evaluation measures

To measure the effectiveness of the proposed model, several classification metrics are applied, such as accuracy, Receiver Operating Characteristic (ROC) curve, precision, Area Under the Curve (AUC), recall, f1-score, and the confusion matrix. Accuracy measures the overall proportion of correctly classified instances. Precision measures the correctness of positive predictions. Recall measures how well the model spots actual positive cases. The F1-score provides a balanced measure by combining recall and precision. The ROC curve displays the balance between correctly identified positives and wrongly flagged negatives across various cutoff points, while the AUC shows the model's discriminative capability. When actual test results are available, a confusion matrix gives a detailed report of a classifier's accuracy. It breaks down every correct and incorrect prediction for a given problem. Based on the classification output, four instances can be derived. They are True Positive (TP), True Negative (TN), False Negative (FN), and False Positive (FP) [34]. These evaluation measures offer key insights into the model's correctness and overall performance, as summarized in Table 4.

3.5. System and algorithm configuration

The complete irrigation prediction framework was implemented on a system featuring 4 GB RAM, a 64-bit architecture, an Intel® Core™ i5–4210 U processor running at 1.70 GHz, a 238 GB SSD, and integrated Intel® HD Graphics. In this study, the IGWO algorithm was set with 60 search agents and run for up to 150 iterations to determine the optimal initial weights for the neural network. After IGWO-based initialization, the artificial neural network was further refined using Adam (lr =0.004) for 100 epochs to enhance convergence.

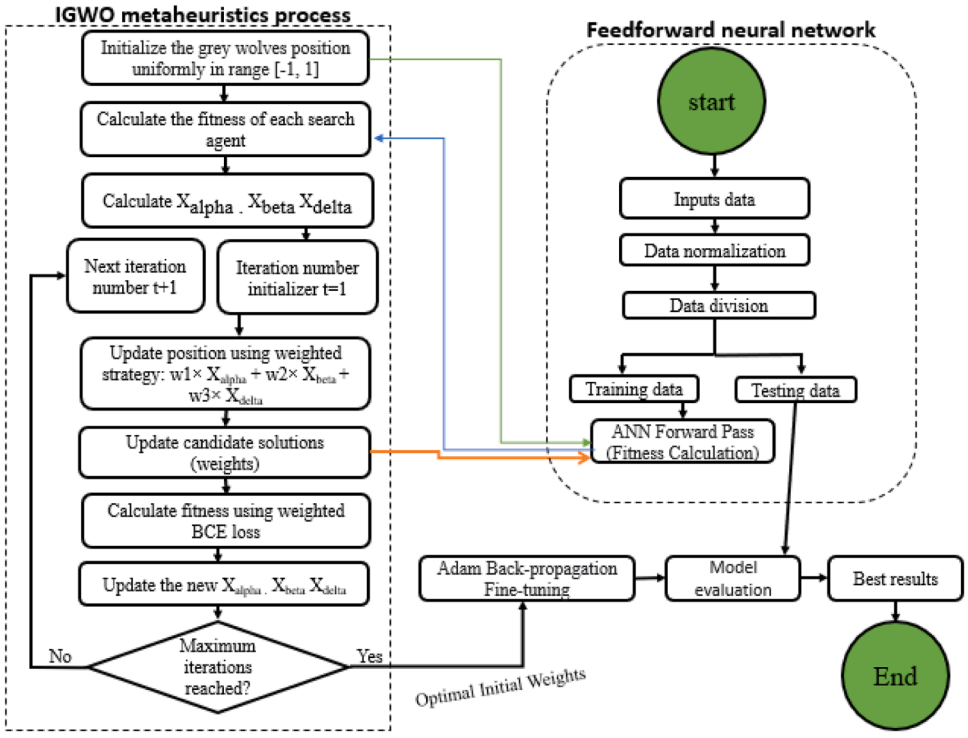


Fig. 4. Structure of IGWO-ANN.

Table 3
Algorithm for IGWO-ANN.

Algorithm for IGWO–ANN

Input: $X \in \mathbb{R}^{n\times 5}$, irrigation class label $y \in \{0,1\}$
Output: Irrigation classification model
Load data
Remove null samples
Scale the data to a specified range
$X \leftarrow data$
$X_{scaled} = \frac{(X - X_{min})}{(X_{max} - X_{min})}$
Parameter Initialization
Number of wolves $N = 60$; Maximum iterations $T = 150$; w1=0.6→0.5, w2=0.3, w3=residual; Position range [−1,1]
for $t = 1 : T$
for $i = 1 : N$
Inject weight vector W_i into ANN
$Fitness_i \leftarrow WeightedBCE(ANN(X_{train}), y_{train}, class_weight = 2.2)$
end for
Identify α, β, δ solutions
for $i = 1 : N$
Update W_i *using using IGWO weighted position update* : $W_i = w1 \times X1 + w2 \times X2 + w3 \times X3$
end for
end for
$W^* \leftarrow W_\alpha$
ANN Training
Train ANN using Adam optimizer with weighted binary cross-entropy loss
Testing
$output \leftarrow ANN(X_{test})$

4. Experiments

4.1. Sensitivity analysis of optimization convergence for different iterations

To assess convergence stability of the proposed approach, a sensitivity analysis was conducted across five iteration configurations of 50, 100, 150, 200, and 300 with five independent runs each. Although the fitness curve exhibits gradual decline without full plateauing by the 150th iteration as shown in Fig. 5 and presented in Table 5. The results confirm that 150 iterations represent the optimal configuration, achieving mean accuracy of 98.52% ± 0.14%, F1-score of 95.84% ± 0.38%, and AUC of 0.9998 ± 0.0000 with the lowest variance across all configurations. Increasing to 200 iterations introduces stochastic instability (accuracy: 97.99% ± 1.20%; F1-score: 94.22% ± 3.52%), while 300 iterations yields marginally lower fitness (0.1113 vs. 0.1125) without statistically significant classification improvement despite higher computational overhead. This behavior is characteristic of metaheuristic optimization on high-dimensional weight spaces where the fitness landscape becomes increasingly flat and yielding diminishing

Table 4
List of performance metrics.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \qquad TPR = \frac{TP}{(TP + FN)}$$

$$Precision = \frac{TP}{(TP + FP)} \qquad FPR = \frac{FP}{(FP + TN)}$$

$$Recall = \frac{TP}{(TP + FN)} \qquad AUC = \int_0^1 TPR(FPR)d(FPR)$$

$$F1 - score = \frac{(2 \times Precision \times Recall)}{(Precision + Recall)}$$

returns. Consequently, 150 iterations were confirmed as the practical convergence threshold maximizing classification stability and computational efficiency, with the ANN weight initialization at this configuration being sufficiently near-optimal for subsequent gradient-based fine-tuning.

4.2. GWO-ANN variants experiments

To strengthen the justification for the selected optimization strategy, a systematic comparison of four GWO-ANN variants was conducted. The compared variants include Chaotic GWO-ANN, Adaptive GWO-ANN, Standard GWO-ANN, and the proposed Improved GWO-ANN (IGWO-ANN), with same ANN architecture and evaluated across 10 independent runs to ensure statistical reliability. The results are presented in Table 6. Chaotic GWO-ANN, which replaces random coefficients with logistic chaotic map values during ANN weight initialization, achieved the lowest performance across all metrics with an accuracy of 95.97% ± 1.55%, precision of 86.42% ± 5.02%, recall of 90.74% ± 4.41%, F1-score of 88.48% ± 4.35%, and AUC of 0.9758 ± 0.0250. The high standard deviations across all metrics confirm that chaotic initialization introduces excessive randomness that destabilizes the ANN weight search process for this dataset, producing unreliable and inconsistent classification outcomes across runs. Adaptive GWO-ANN, which employs a nonlinear decrease of the control parameter to extend exploration before ANN weight initialization, achieved a competitive accuracy of 98.21% ± 0.60% and AUC of 0.9984 ± 0.0016, but lower precision of 92.19% ± 1.27% and F1-score of 94.89% ± 1.77% compared to Standard GWO-ANN and IGWO-ANN, suggesting that extended exploration alone does not sufficiently improve the quality of ANN weight initialization for this irrigation classification task. Standard GWO-ANN achieved 98.65% ± 0.54% accuracy, precision of 93.02% ± 2.55%, recall of 99.59% ± 0.76%, F1-score of 96.18% ± 1.51%, and AUC of 0.9993 ± 0.0006, demonstrating strong and consistent baseline performance for ANN weight initialization. The proposed IGWO-ANN, which assigns adaptive influence weights to the alpha, beta, and delta leaders through a weighted position update strategy for ANN parameter initialization, consistently outperformed all variants across every metric, achieving the highest accuracy of 98.72% ± 0.44%, precision of 93.46% ± 1.32%, recall of 99.43% ± 1.47%, F1-score of 96.35% ± 1.25%, and AUC of 0.9994 ± 0.0009. The performance advantage of

Table 5
Performance comparison of IGWO-ANN across different iteration.

Iterations	Accuracy	F1-Score	AUC	Final Fitness
50	98.27±0.26%	95.10±0.80%	0.9990 ±0.0007	0.1839 ±0.0250
100	98.24±0.43%	95.01±1.24%	0.9989 ±0.0009	0.1394 ±0.0345
150	**98.52 ±0.14%**	**95.84 ±0.38%**	**0.9998 ±0.0000**	**0.1125 ±0.0162**
200	97.99±1.20%	94.22±3.52%	0.9974 ±0.0032	0.1508 ±0.0502
300	98.77±0.30%	96.51±0.82%	0.9996 ±0.0004	0.1113 ±0.0242

Table 6
Comparison of different GWO variants.

Variant	Accuracy	Precision	Recall	F1-Score	AUC
Chaotic GWO-ANN	95.97% ± 1.55%	86.42% ± 5.02%	90.74% ± 4.41%	88.48% ± 4.35%	0.9758 ± 0.0250
Adaptive GWO-ANN	98.21% ± 0.60%	92.19% ± 1.27%	97.79% ± 2.74%	94.89% ± 1.77%	0.9984 ± 0.0016
Standard GWO-ANN	98.65% ± 0.54%	93.02% ± 2.55%	99.59% ± 0.76%	96.18% ± 1.51%	0.9993 ± 0.0006
IGWO-ANN	**98.72% ± 0.44%**	**93.46% ± 1.32%**	**99.43% ± 1.47%**	**96.35% ± 1.25%**	**0.9994 ± 0.0009**

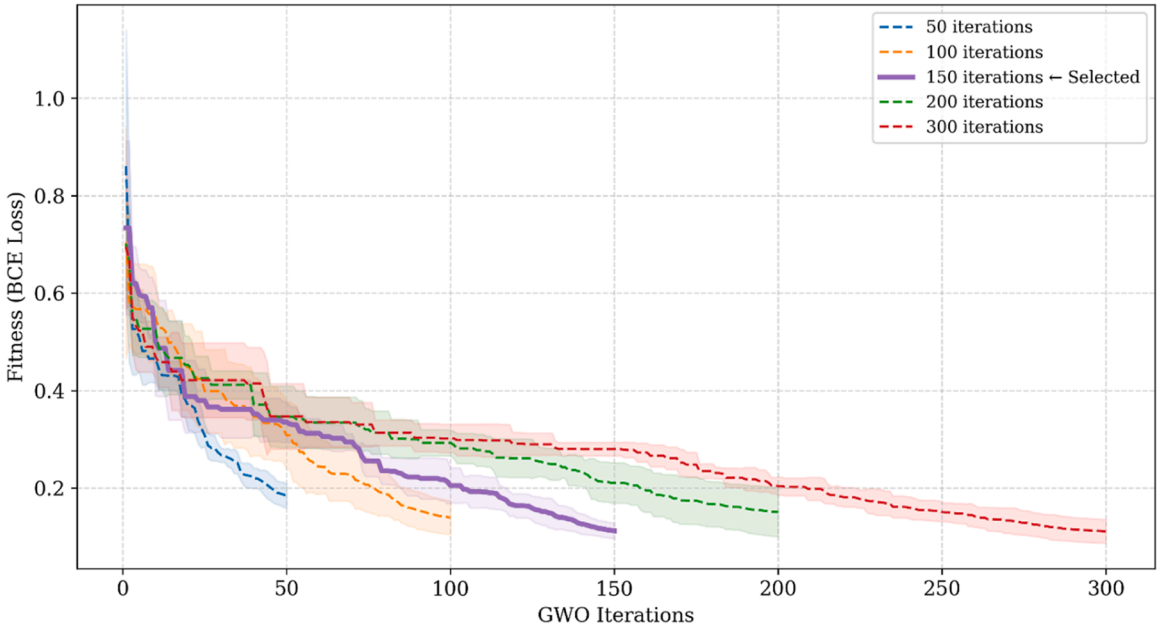


Fig. 5. IGWO-ANN convergence curves across different iteration.

IGWO-ANN over all other variants is primarily attributed to its weighted position update mechanism, which gives greater influence to the best solution during ANN weight initialization, resulting in more consistent near-optimal parameter configurations for the subsequent gradient-based fine-tuning phase. The consistently lowest standard deviation across all metrics confirms that IGWO-ANN provides the most reliable and reproducible ANN initialization quality across all independent runs, making it the most suitable optimization strategy for the proposed irrigation classification framework.

5. Results and discussion

This section discusses the specific outcomes obtained from the IGWO-ANN prediction unit, and a comparative analysis clearly demonstrates the performance of the IGWO-ANN in irrigation classification. The findings show how the implemented model achieves notable performance in predicting irrigation needs. To properly train and validate the model, five input variables are considered: crop type, soil moisture, air temperature, air humidity, and soil temperature. The dataset was divided into training (80%) and test (20%) sets using a fixed random state of 42. All models were executed across 10 independent runs and results are reported as mean ± standard deviation to ensure statistical reliability. Initially, IGWO with a population of 60 wolves and 150 maximum iterations is employed to obtain optimal initial weights for the ANN by minimizing the weighted Binary Cross-Entropy loss through a forward pass on the training data. Afterward, the network is fine-tuned using Adam-based backpropagation with a learning rate of 0.004 over 100 epochs. The dataset utilized in this study exhibits a notable class imbalance, with 2994 samples belonging to the no-irrigation class and 590 samples belonging to the irrigation class, yielding an imbalance ratio of approximately 5:1. The minority class (irrigation) therefore represents only 16.5% of the total 3584 samples, which constitutes a moderately severe imbalance that can bias the model toward the majority class if left unaddressed. To mitigate this issue, a weighted Binary Cross-Entropy loss function was employed during the gradient-based optimization phase. Specifically, a class weight value of 2.2 was assigned to the minority class, penalizing misclassifications of irrigation samples more heavily than those of the no-irrigation class. This weighting strategy encourages the model to pay greater attention to the underrepresented class during parameter updates, thereby promoting more balanced classification performance across both classes. The experimental results indicate that the integration of IGWO-based weight initialization with gradient-based fine-tuning contributes to stable learning behavior and improved classification performance. It is acknowledged that a comparative analysis of alternative imbalance handling techniques, such as SMOTE or undersampling strategies, was not conducted in this study. A systematic comparison of such methods represents a direction for future investigation to further validate the effectiveness of the adopted weighting strategy.

5.1. Comparative methods

To evaluate the effectiveness of the proposed IGWO-ANN model, its performance is compared with a comprehensive set of baseline and state-of-the-art models including SVM [35], standard ANN, TanElu [3], XGBoost, Random Forest, LSTM, CNN-LSTM, Transformer, PSO-ANN, ABC-ANN and GA-ANN. Additionally, a comparison across four GWO variants namely, Standard GWO, Chaotic GWO, Adaptive GWO, and IGWO was conducted to provide empirical justification for the selected optimization strategy. The comparison is conducted using standard performance metrics, namely accuracy, precision, recall, F1-score, AUC, and training time. To ensure a fair and controlled comparison, all neural network models are developed using the same dataset, identical 80:20 train/test split with random state of 42, and the same weighted Binary Cross-Entropy loss function. PSO-ANN, ABC-ANN and GA-ANN share the exact same ANN architecture as the proposed IGWO-ANN model ensuring that any performance differences are attributable exclusively to the optimization algorithm rather than variations in architecture, data, or training configuration. All models were executed across 10 independent runs and results are reported as mean ± standard deviation to ensure statistical reliability and reproducibility.

5.2. Comparative analysis

This section illustrates a comparative evaluation of the approaches used in the classification of irrigation using conventional techniques and the proposed model.

5.2.1. Results of IGWO-ANN for irrigation classification

The most notable outcome of this study is the strong performance achieved by the proposed IGWO-ANN model, which attains a mean accuracy of 98.72% ± 0.44% across 10 independent runs, with a single best run achieving 98.88% accuracy. This accomplishment is accompanied by commendable metrics including recall, F1-score, and precision of 100.00%, 96.83%, and 93.85%, respectively, demonstrating particularly strong sensitivity in correctly identifying all irrigation requirements without any false negatives. Fig. 6 illustrates the evolution of accuracy and loss across the 100 training epochs for the proposed IGWO-ANN model. The accuracy curve demonstrates a stable and consistent upward trend for both training and test data from the early epochs, with both curves converging toward approximately 99% by the final epoch. The close alignment between training and test accuracy throughout all epochs confirms that the model generalizes well to unseen data without overfitting. In terms of loss behavior, both training loss and validation loss exhibit a rapid reduction during the early epochs, followed by smooth and consistent convergence toward the global optimum. Notably, the validation loss remains slightly lower than the training loss throughout the fine-tuning phase. This behavior is a direct and expected consequence of the weighted Binary Cross-Entropy loss function, which artificially inflates the training loss by applying heavier penalties on minority class misclassifications during parameter updates. Since this penalty is applied during training but not during test evaluation, the validation loss naturally appears lower, confirming that this observation reflects the intended class imbalance handling strategy rather than any form of underfitting or data anomaly. The gradient-based fine-tuning phase required only 0.44 s, demonstrating exceptional computational efficiency as a direct result of the IGWO-based weight initialization providing near-optimal starting parameters that enable rapid convergence with minimal gradient updates. This performance improvement can be attributed to the synergistic integration of IGWO-based optimized initialization and the class-weighted loss function. The observed performance advantage of the proposed model is primarily attributed to the IGWO's weighted position update strategy, which assigns greater influence to the alpha wolf during parameter initialization, resulting in more consistent and near-optimal weight configurations that enable faster and more stable convergence during the subsequent gradient-based fine-tuning phase. Additionally, the class-weighted loss function is critical for overall model performance as it specifically addresses the 5:1 class imbalance to maximize minority class recall, achieving perfect recall of 100.00% in the best run. Furthermore, because the network architecture and Adam optimizer were held strictly constant across all experimental configurations, the observed gains are definitively driven by these targeted training strategies rather than architectural design choices or optimizer selection. Ultimately, this synergistic optimization improves the model's ability to capture complex nonlinear patterns in the IoT-collected agricultural data that may not be efficiently learned through conventional random initialization approaches.

The confusion matrix in Fig. 7 shows excellent model performance, as the model misclassifies only 8 records. The high number of correct classifications for both the irrigation and no-irrigation classes indicates that the model does not exhibit bias toward any specific class.

The proposed model was evaluated against a comprehensive set of

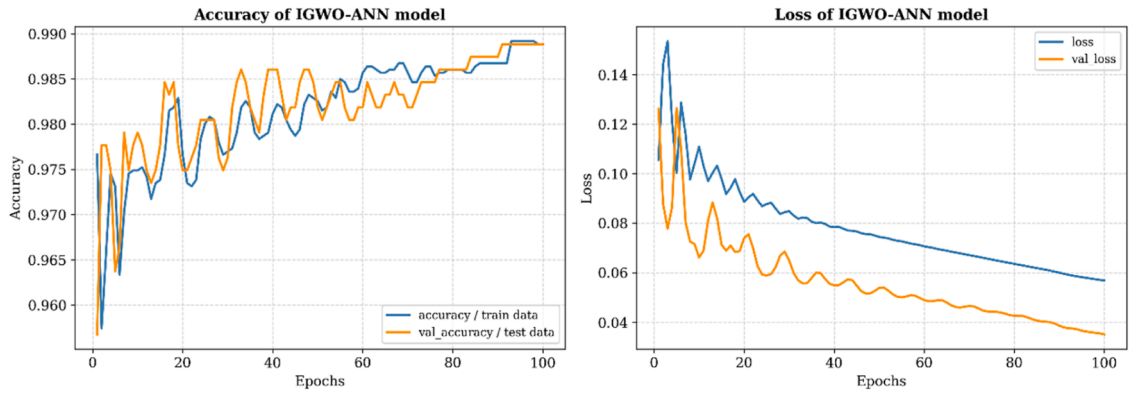


Fig. 6. Accuracy of the proposed IGWO-ANN.

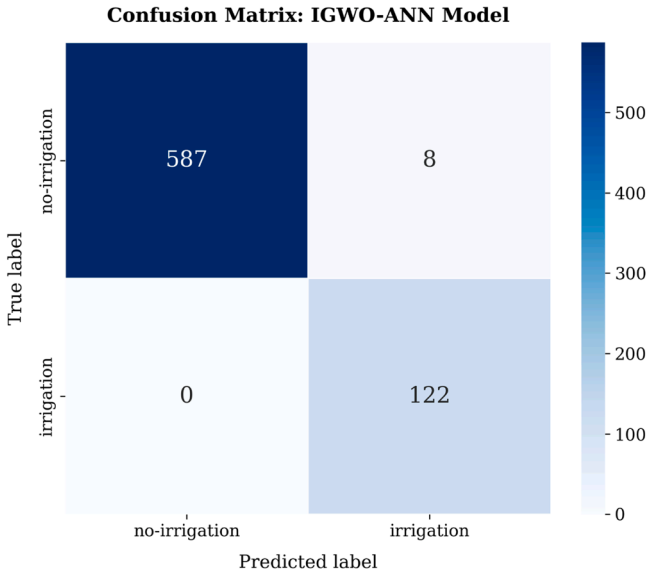


Fig. 7. Confusion Matrix of IGWO-ANN.

baseline and state-of-the-art approaches, including SVM, standard ANN, TanElu, XGBoost, Random Forest, LSTM, CNN-LSTM, Transformer, PSO-ANN, and GA-ANN, to provide a rigorous benchmark comparison. Table 7 shows that the IGWO-ANN framework has the highest mean balanced classification performance among all hybrid and deep learning models, attaining 98.72% ± 0.44% accuracy, 93.46% ± 1.32% precision, 99.43% ± 1.47% recall, 96.35% ± 1.25%, F1-score, and 0.9994 ± 0.0009 AUC. Among the compared models, Random Forest (RF) and XGBoost achieved 100% and 99.86% accuracy respectively. Despite

Table 7
Comparison metrics of IGWO-ANN and Other Methods.

Model	Accuracy	Precision	Recall	F1-score
SVM	87.29%	86.77%	85.42%	86.09%
ANN	96.38%	93.75%	83.80%	88.50%
TanElu	98.24%	92.11%	97.77%	94.85%
XGBoost	99.86%	99.19%	100.00%	99.59%
RF	100.00%	100.00%	100.00%	100.00%
LSTM	94.56%	83.20%	85.25%	84.21%
CNN-LSTM	98.19%	90.98%	99.18%	94.90%
PSO-ANN	98.08%	91.88%	97.30%	94.50%
GA-ANN	98.24%	92.67%	97.38%	94.94%
ABC-ANN	98.09%	91.82%	97.46%	94.54%
Transformer	97.77%	97.32%	89.34%	93.16%
IGWO-ANN	98.72± 0.44%	93.46 ± 1.32%	99.43% ± 1.47%	96.35% ± 1.25%

aggressive regularization applied to both models, RF consistently achieved perfect classification, indicating that the dataset contains naturally well-separable feature boundaries that tree-based ensemble methods exploit through axis-aligned splitting strategies. However, such perfect performance on a small, single-location, two-crop dataset raises concerns regarding generalizability to unseen agricultural environments, diverse soil conditions, and broader crop varieties. Furthermore, tree-based models such as Random Forest and XGBoost do not provide an interpretable optimization trajectory and lack the adaptive weight refinement mechanism that the proposed IGWO-ANN offers for evolving agricultural data. After RF and XGBoost, the accuracy of the models decreased progressively. TanElu achieved 98.24% [3], followed by GA-ANN at 98.24%, PSO-ANN at 98.08%, CNN-LSTM at 98.19%, ABC-ANN at 98.09%, standard ANN at 96.38% [3], Transformer at 97.77%, LSTM at 94.56%, and SVM at 87.29% [35], respectively. The enhancement in achieved accuracy of the proposed IGWO-ANN over all hybrid optimization models is particularly noteworthy, as PSO-ANN, GA-ANN, and ABC-ANN share an identical ANN architecture with the same training configuration applied in this study, where ANN weights are initialized using PSO velocity-based updates, GA crossover and mutation operations, and ABC employed bee foraging behavior respectively, before gradient-based fine-tuning. The only distinction lies in the metaheuristic optimization algorithm used for weight initialization, which isolates the contribution of IGWO and confirms that its hierarchical hunting mechanism guided by adaptive influence weights assigned to alpha, beta, and delta wolves provides superior and more consistent weight space exploration compared to all alternative metaheuristic strategies. The underlying aim of incorporating IGWO is to harness the combined virtues of both global exploration and local refinement. The employment of IGWO lends the benefit of searching the vast parameter space to determine the optimal initial weights and biases, effectively bypassing the local minima traps that often hinder standard gradient-based initialization in models like ANN and SVM. Conversely, the Adam backpropagation algorithm comes into play as a local refiner, facilitating expedited learning and precise convergence to the global optimum once the IGWO has narrowed the search space. Moreover, the dataset encompasses intricate, real-time nonlinear relationships among its data, and in this context the hybrid IGWO-ANN model manages complex data patterns more effectively than standalone models while maintaining a computationally efficient training process.

Fig. 8 shows the ROC curve for the proposed IGWO-ANN framework, evaluated across 10 independent runs to ensure statistical reliability. The model achieves a mean AUC of 0.9994 ± 0.0009, reflecting both exceptional discriminative capability and highly consistent performance across all runs. The negligible standard deviation in AUC confirms that the proposed IGWO-ANN produces stable and reproducible classification results rather than a single favorable outcome. It is observed that the mean ROC curve climbs steeply toward the upper-left corner across

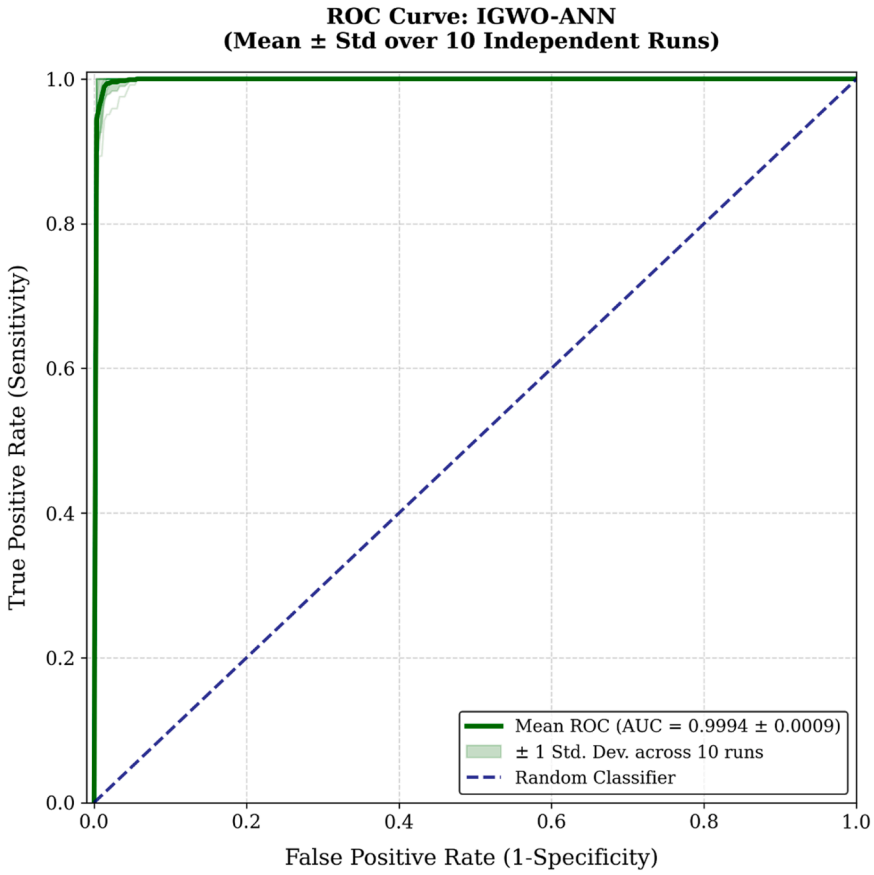


Fig. 8. ROC curve for IGWO-ANN.

all 10 runs, depicting that the model consistently maintains a high True Positive Rate (TPR/Sensitivity) while keeping the False Positive Rate (FPR/1-Specificity) at a minimum. The narrow shaded variance band surrounding the mean curve further demonstrates that the model's class separability is highly robust to random initialization differences between runs, with individual run AUC values ranging from 0.9971 to 1.0000. This performance is attributed to the hybrid IGWO-ANN strategy, which combines the weighted position update global search optimization with gradient-based local refinement. The adaptive influence weighting of the alpha, beta, and delta wolves during the global search phase enables the network to consistently identify near-optimal initial weight configurations, which in turn allows the gradient-based fine-

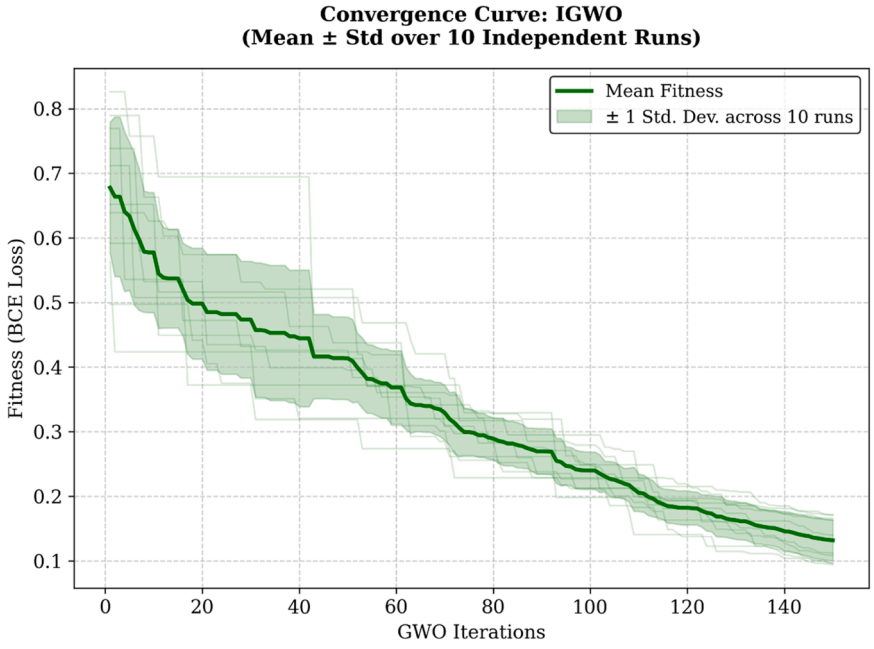


Fig. 9. Convergence curve of ANN-IGWO.

tuning phase to achieve a high degree of class separability and robustness, effectively capturing the complex nonlinear thresholds present in the agricultural IoT dataset.

Fig. 9 illustrates the convergence curve of the IGWO during the global search phase of the hybrid approach, evaluated across 10 independent runs to provide a statistically grounded analysis. The mean fitness curve exhibits a steep initial decline from approximately 0.68, followed by a consistent and progressive reduction reaching a final mean value of approximately 0.13 at the 150th iteration. The shaded variance band representing ± 1 standard deviation across all 10 runs demonstrates that while individual runs exhibit some variation during early iterations due to stochastic population initialization, the convergence trajectories consistently stabilize and the variance band progressively narrows as iterations advance, confirming the reliability and reproducibility of the IGWO search process across all independent runs.

5.2.2. *Comparison of execution time between the proposed IGWO-ANN and other methods on the same dataset*

Measuring execution time in smart watering code is essential, as it provides insight into the computational efficiency of different methods. This evaluation helps identify operations that contribute to increased processing time and enables a fair comparison of the proposed IGWO–ANN with other approaches in terms of performance efficiency. Based on Fig. 10, it is evident that the proposed IGWO-ANN required only 0.44 s for the backpropagation fine-tuning phase in irrigation classification. It is important to clarify that this measurement reflects exclusively the gradient-based optimization phase, wherein the Adam optimizer refines the GWO-initialized weights over 100 epochs. The IGWO-based global search phase, which involves 150 iterations with 60 wolves, incurs additional computational overhead that is separate from this measurement. Among all compared models, XGBoost achieved the shortest total training time of 0.13 s, followed by Random Forest at 0.78 s. However, as discussed in the model comparison section, these tree-based models exploit simple decision boundaries on this dataset and do not involve the iterative weight optimization process that characterizes neural network-based approaches. Among all neural network and hybrid optimization models, the PSO-ANN at 17.85 s, GA-ANN at 13.93 s, Transformer at 10.45 s, CNN-LSTM at 26.95 s, and LSTM at 2.87 s. The proposed approach duration is also significantly shorter when compared to TanElu and the standard ANN, which required 7.88 s and 7.92 s respectively, both of which were measured over their complete training cycles. The reason for this efficiency lies in the two-stage optimization strategy of the proposed model. During the IGWO-based global search phase, IGWO performs a metaheuristic search to identify near-optimal weights and biases for the ANN, effectively positioning the model parameters in a highly promising region of the search space. Consequently, during the backpropagation fine-tuning phase, the gradient-based optimizer starts from these near-optimal initial parameters rather than random initialization, allowing it to converge rapidly with minimal updates and significantly reduced computational effort. In contrast, hybrid models such as PSO-ANN and GA-ANN, despite sharing the same architecture, require substantially longer fine-tuning durations because their respective optimization algorithms are less effective at identifying promising initial weight configurations, forcing the subsequent gradient-based phase to perform more extensive parameter updates. Similarly, conventional models such as standard ANN and TanElu rely on random weight initialization and must independently explore the error surface using gradient-based optimization, which increases computational complexity and prolongs convergence time, particularly when escaping local minima.

In summation, the study vividly portrays the significance of the IGWO-ANN hybrid framework in augmenting machine learning model performance. It substantiates the framework's potency in capturing intricate data patterns, optimizing initial parameters to avoid local minima, and enhancing model accuracy and stability. The study's results highlight the critical role of metaheuristic-based weight initialization in crafting effective neural network architectures tailored to complex datasets and real-time scenarios.

6. Conclusion and future work

This research demonstrates that the integration of the Improved Grey Wolf Optimization with an Artificial Neural Network enhances the optimization capability of the model by improving the update process of ANN weights and biases through an adaptive weighted position update strategy that assigns greater influence to the alpha wolf during parameter initialization. The proposed IGWO-ANN model outperforms traditional deep learning, machine learning, and hybrid optimization approaches in classifying irrigation requirements for different crops. Evaluation across various metrics, including accuracy, f1-score, ROC curve, precision, recall, AUC, confusion matrix and training execution time, underscores the reliability and robustness of the proposed model. The proposed IGWO-ANN model attains a mean accuracy of 98.72% ± 0.44%, AUC of 0.9994 ± 0.0009, recall of 99.43% ± 1.47%, F1-score of 96.35% ± 1.25%, and precision of 93.46% ± 1.32%, demonstrating both exceptional classification performance and high statistical stability

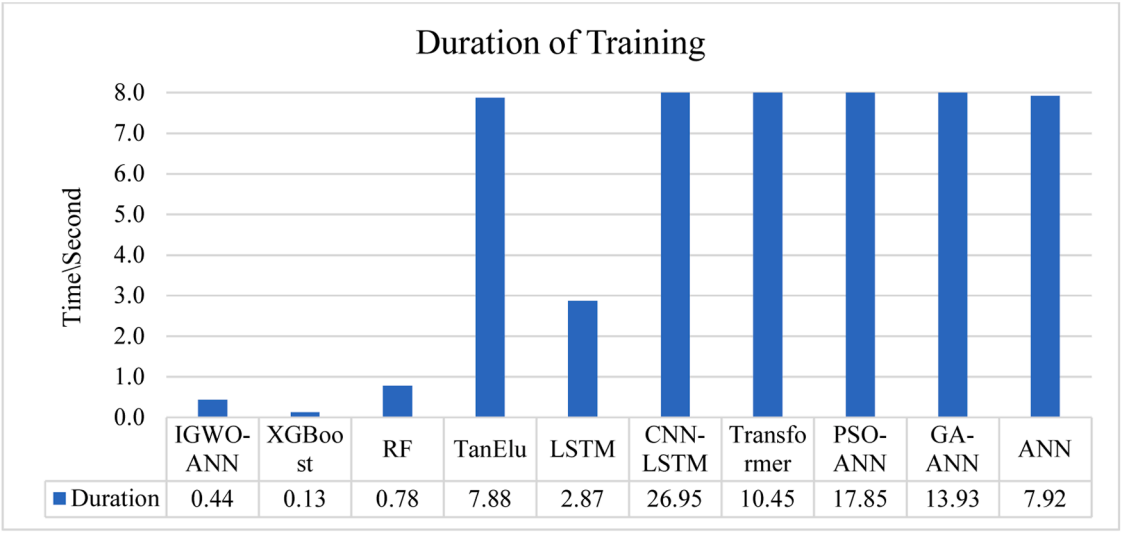


Fig. 10. Execution time for irrigation classification.

across runs. The gradient-based fine-tuning phase required only 0.44 s, demonstrating computational efficiency as a direct result of IGWO-based weight initialization providing near-optimal starting parameters. To justify the selection of IGWO over other optimization strategies, a systematic comparison of four GWO-ANN variants was conducted, namely Chaotic GWO-ANN, Adaptive GWO-ANN, Standard GWO-ANN, and IGWO-ANN. The results reveal that IGWO-ANN consistently achieves the highest performance and lowest variance across all metrics, validating the weighted position update mechanism as the most effective initialization strategy for this irrigation classification task. The implications of this research extend beyond academic contributions and offer potential benefits for agricultural stakeholders. The proposed approach can assist farmers in improving crop yield, lowering operational costs, reducing water wastage, and enhancing crop quality through more efficient and data-driven irrigation management. Despite these outstanding findings, certain limitations remain. The dataset utilized in this research was collected over a one-week period and encompasses only two crop types, which may limit the generalizability of the proposed model across broader agricultural scenarios involving different seasons, climates, soil conditions, and crop varieties. Furthermore, external validation using independent datasets from diverse geographical regions was not performed in this study, as such data was not available at the time of this research. From an IoT deployment perspective, practical considerations such as sensor noise robustness, missing sensor readings beyond simple exclusion, deployment scalability across large heterogeneous farm networks, edge-device implementation, power consumption constraints, and real-time inference requirements were not empirically investigated, as this study utilized a secondary dataset collected from existing IoT sensors. A comparative analysis of alternative class imbalance handling techniques such as SMOTE or undersampling strategies was not conducted, representing an acknowledged boundary of the current work. To address these limitations, future work will investigate the collection of larger and more diverse datasets encompassing multiple crop types, different seasons, varying soil types, and distinct geographic locations to improve the generalizability and robustness of the proposed framework. A systematic comparison of alternative imbalance handling techniques including SMOTE, undersampling, and hybrid sampling strategies will be conducted to further validate the effectiveness of the adopted weighting strategy. Additionally, the proposed IGWO-ANN framework will be evaluated for deployment on resource-constrained edge devices, with particular attention to real-time inference performance, power efficiency, and robustness to sensor noise and missing data under actual field conditions. Such expansion will improve the robustness and adaptability of the proposed model in real-world deployments, and the system will be validated through real-world field trials to assess its performance and practical applicability under actual farming conditions.

CRediT authorship contribution statement

Zameer Ahmad: Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation, Conceptualization. **Mohd Adib Omar:** Writing – original draft, Visualization, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Saqib Jamil:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nibras Abdullah:** Writing – review & editing, Visualization, Validation, Software, Resources, Formal analysis, Conceptualization. **Sk. A. Shezan:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Naveed Ahmad:** Writing – review & editing, Visualization, Validation, Software, Resources, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors extend their appreciation to the Deanship of Scientific Research at Northern Border University, Arar, KSA for funding this research work Through the project number “NBU-FFR-2026-1902-07".

Data availability

Data will be shared with reasonable request.

References

[1] O. Dubois, The State of the World’s Land and Water Resources for Food and Agriculture: Managing Systems at Risk, London Earthscan, 2011, p. 285, xxiii +.
[2] L. Colizzi, G. Dimauro, E. Guerriero, N. Lomonte, Artificial intelligence and IoT for water saving in agriculture: a systematic review, Smart Agric. Technol. 11 (2025) 101008.
[3] B.H. Jawad, O.A. Alwesabi, N. Abdullah, A.A. Mohammed, Hybrid artificial neural network activation function to reduce water wastage in agricultural irrigation, IEEe Access. 13 (2025) 93302–93322.
[4] A. Sharma, A. Jain, P. Gupta, V. Chowdary, Machine learning applications for precision agriculture: a comprehensive review, IEEe Access. 9 (2020) 4843–4873.
[5] L. Preite, G. Vignali, Artificial intelligence to optimize water consumption in agriculture: a predictive algorithm-based irrigation management system, Comput. Electron. Agric. 223 (2024) 109126.
[6] S. Sadeghi Tabas, S. Samadi, Variational bayesian dropout with a gaussian prior for recurrent neural networks application in rainfall–runoff modeling, Environ. Res. Lett. 17 (6) (2022) 065012.
[7] L. Windheuser, R. Karanjit, R. Pally, S. Samadi, N. Hubig, An end-to-end flood stage prediction system using deep neural networks, Earth Space Sci. 10 (1) (2023) e2022EA002385.
[8] L. Umutoni, V. Samadi, Application of machine learning approaches in supporting irrigation decision making: a review, Agric. Water. Manage 294 (2024) 108710.
[9] K. Hemalatha, K.U. Rani, Advancements in multi-layer perceptron training to improve classification accuracy, Int. J. Recent Innov. Trends Comput. Commun. 5 (6) (2017) 353–357.
[10] A. Mosavi, et al., Predicting soil electrical conductivity using multi-layer perceptron integrated with grey wolf optimizer, J. Geochem. Explor. 220 (2021) 106639.
[11] A.T. Maadapoosi, V. Balamurugan, V. Vedanarayanan, S.A. Nisha, R. Narmadha, Grey wolf optimization-based artificial neural network in the development of an automated heart disease prediction model, Int. J. Fuzzy Logic Intell. Syst. 24 (3) (2024) 231–241.
[12] R.Y. Mujawar, R. Lakshminarayanan, A. Jyothi, S. Patnaik, K. Dhayalini, IoT-enabled intelligent irrigation system with machine learning-based monitoring, for effective rice cultivation, Int. J. Intell. Syst. Appl. Eng. 12 (11s) (2024) 557–565.
[13] D.T. Santosh, et al., Development of IoT based intelligent irrigation system using particle swarm optimization and XGBoost techniques, Bull. Electric. Eng. Inform. 13 (3) (2024) 1927–1934.
[14] S. Maroufpoor, E. Maroufpoor, O. Bozorg-Haddad, J. Shiri, Z.M. Yaseen, Soil moisture simulation using hybrid artificial intelligent model: hybridization of adaptive neuro fuzzy inference system with grey wolf optimizer algorithm, J. Hydrol. 575 (2019) 544–556.
[15] B. He, B. Jia, Y. Zhao, X. Wang, M. Wei, R. Dietzel, Estimate soil moisture of maize by combining support vector machine and chaotic whale optimization algorithm, Agric. Water. Manag. 267 (2022) 107618.
[16] R. Moazenzadeh, B. Mohammadi, M.J.S. Safari, K.-w. Chau, Soil moisture estimation using novel bio-inspired soft computing approaches, Eng. Appl. Computat. Fluid Mech. 16 (1) (2022) 826–840.
[17] P. Majumdar, D. Bhattacharya, S. Mitra, R. Solgi, D. Oliva, B. Bhusan, Demand prediction of rice growth stage-wise irrigation water requirement and fertilizer using Bayesian genetic algorithm and random forest for yield enhancement, Paddy Water Environ. 21 (2) (2023) 275–293.
[18] B. Mohammadi, S. Mehdizadeh, Modeling daily reference evapotranspiration via a novel approach based on support vector regression coupled with whale optimization algorithm, Agric. Water. Manag. 237 (2020) 106145.
[19] M.Y. Chia, Y.F. Huang, C.H. Koo, Swarm-based optimization as stochastic training strategy for estimation of reference evapotranspiration using extreme learning machine, Agric. Water. Manage 243 (2021) 106447.
[20] S. Chatterjee, N. Dey, S. Sen, Soil moisture quantity prediction using optimized neural supported model for sustainable agricultural applications, Sustain. Comput.: Inform. Syst. 28 (2020) 100279.
[21] S. Mirjalili, S.M. Mirjalili, A. Lewis, Grey wolf optimizer, Adv. Eng. Softw. 69 (2014) 46–61.

[22] M.N. Amar, N. Zeraibi, K. Redouane, Bottom hole pressure estimation using hybridization neural networks and grey wolves optimization, Petroleum 4 (4) (2018) 419–429.
[23] A. Behnood, E.M. Golafshani, Predicting the compressive strength of silica fume concrete using hybrid artificial neural network with multi-objective grey wolves, J. Clean. Prod. 202 (2018) 54–64.
[24] P. Majumdar, S. Mitra, D. Bhattacharya, Soil moisture simulation of rice using optimized Support Vector Machine for sustainable agricultural applications, Sustain. Comput.: Inform. Syst. 40 (2023) 100924.
[25] M. Bidabadi, H. Babazadeh, J. Shiri, A. Saremi, Estimation reference crop evapotranspiration (ET0) using artificial intelligence model in an arid climate with external data, Appl. Water. Sci. 14 (1) (2024) 3.
[26] Y. Tikhamarine, A. Malik, A. Kumar, D. Souag-Gamane, O. Kisi, Estimation of monthly reference evapotranspiration using novel hybrid machine learning approaches, Hydrol. Sci. J. 64 (15) (2019) 1824–1842.
[27] S. Maroufpoor, O. Bozorg-Haddad, E. Maroufpoor, Reference evapotranspiration estimating based on optimal input combination and hybrid artificial intelligent model: hybridization of artificial neural network with grey wolf optimizer algorithm, J. Hydrol. 588 (2020) 125060.
[28] S. Nosratabadi, K. Szell, B. Beszedes, F. Imre, S. Ardabili, A. Mosavi, Comparative analysis of ANN-ICA and ANN-GWO for crop yield prediction, in: 2020 RIVF International Conference on Computing and Communication Technologies (RIVF), IEEE, 2020, pp. 1–5.
[29] K. Shahverdi, J. Maestre, Gray wolf optimization for scheduling irrigation water, J. Irrig. Drain. Eng. 148 (7) (2022) 04022020.
[30] P. Soni, R. Kumar, S. Mishra, S. Swain, P. Mishra, S.S. Kumar, Optimized neural network for soil moisture prediction in precision agriculture, Measurement 252 (2025) 117380.
[31] A. Gjecka, M. Fetaji, Hybrid machine learning method for prediction of thyroid diseases using GWO-ANN, Int. J. Latest Trans. Eng. Sci. (IJLTES) 15 (2) (2024).
[32] H. Faris, I. Aljarah, M.A. Al-Betar, S. Mirjalili, Grey wolf optimizer: a review of recent variants and applications, Neural Comput. Appl. 30 (2) (2018) 413–435.
[33] D.D. Nguyen, et al., Bagging and multilayer perceptron hybrid intelligence models predicting the swelling potential of soil, Transp. Geotechn. 36 (2022) 100797.
[34] R. Patil, S. Tamane, A comparative analysis on the evaluation of classification algorithms in the prediction of diabetes, Int. J. Electric. Comput. Eng. 8 (5) (2018) 3966.
[35] A. Bhoi, et al., IoT-IIRS: internet of things based intelligent-irrigation recommendation system using machine learning approach for efficient water usage, PeerJ Comput. Sci. 7 (2021) e578.